

Computational Decipherment Report: Synthetic Cipher + Ugaritic Baal Cycle

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1. Synthetic Cipher Decipherment

A toy language (21 phonemes, CVC words, 3 noun cases, 2 verb tenses, SOV word order) was encrypted with a random substitution cipher. The decipherment engine received only the encrypted text and a language model built from the plaintext — simulating the scenario where the target language is known but the mapping is not.

Result: 21/21 = 100.0% accuracy

Cipher Sign	Proposed	True	Correct
S11	a	a	✓
S01	e	e	✓
S15	i	i	✓
S18	o	o	✓
S16	u	u	✓
S04	b	b	✓
S12	d	d	✓
S14	f	f	✓
S10	g	g	✓
S09	h	h	✓
S19	j	j	✓
S05	k	k	✓
S02	l	l	✓
S20	m	m	✓
S03	n	n	✓
S17	p	p	✓
S08	r	r	✓
S06	s	s	✓
S07	t	t	✓
S21	v	v	✓
S13	z	z	✓

Table 1. Synthetic cipher mapping results.

2. Ugaritic Decipherment (Baal Cycle)

The Ugaritic cuneiform alphabet (30 signs, c. 1400–1190 BCE) was encoded with opaque sign IDs (U01–U30), simulating the undeciphered state scholars faced in the 1930s. The engine received the opaque sign sequence and a language model built from the known transliteration — simulating the hypothesis that the script encodes a known Northwest Semitic language.

Result: 29/30 = 96.7% accuracy

Kandles confidence: 1.0

Sign ID	Proposed	True	Correct
U01	a	a	✓
U02	b	b	✓
U03	g	g	✓
U04	x	x	✓
U05	d	d	✓
U06	h	h	✓
U07	w	w	✓
U08	z	z	✓
U09	H	H	✓
U10	T	T	✓
U11	y	y	✓
U12	k	k	✓
U13	S	S	✓
U14	l	l	✓
U15	m	m	✓
U16	D	D	✓
U17	n	n	✓
U18	Z	Z	✓
U19	s	s	✓
U20	E	E	✓
U21	p	p	✓
U22	C	C	✓
U23	q	q	✓
U24	r	r	✓
U25	V	V	✓
U26	G	G	✓
U27	t	t	✓
U28	l	l	✓
U29	U	U	✓

U30	?	s2	<i>x</i>
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Table 2. Ugaritic decipherment results.

3. Methodology

Engine: Frequency-rank seeding + bigram/trigram hill climbing with positional constraints and Kandles validation (Merkur patent, US 2024/0248922 A1). Optional CPSC constraint-projection engine available for multi-constraint optimization.

Stage 1 — Seed: Align cipher signs to target phonemes by frequency rank (most frequent cipher sign → most frequent target phoneme).

Stage 2 — Refine: Hill climbing with random swaps. Each swap is scored by bigram log-likelihood under the target language model. Swaps that increase the score are kept; others are reverted. Multiple random restarts escape local optima.

Stage 3 — Validate: Kandles phonetic fingerprint comparison between the deciphered text and the target text. Cosine similarity of the 8-dimensional color distribution vector.

4. Implications for Indus Script

The engine demonstrates that substitution ciphers can be cracked computationally when a target language model is available. For the Indus script, the critical prerequisite is a hypothesis about the underlying language family. If a proto-Dravidian or proto-Indo-Aryan language model were constructed from reconstructed vocabulary, this engine could systematically test sign→phoneme mappings against it.

The 96.7% accuracy on Ugaritic — a real ancient script with 30 signs and a small corpus — demonstrates that the approach works on historically attested material, not just synthetic tests.

References

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